

Every represented operator is compact

[Back to Project mathematical routes](#)

Scope

The preimage ideal of compact operators and a separating character exclude a noncompact represented image. The contradiction uses the same projection while forcing the character to take both zero and one.

7 direct Cards + 14 reused prerequisites = 21 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [Every operator in a singleton image is compact](#)
- Key definition: [The preimage ideal of the compact operators](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#) [3](#) [4](#) [5](#) [6](#) [7](#) [8](#)

Level 0 (6 Cards)

Level 0

Maximal abelian *-subalgebras

Defines maximality by inclusion among commutative unital *-subalgebras.

`MathlibAnnex.Analysis.CStarAlgebra.IsMaximalAbelian`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A maximal abelian subalgebra containing a self-adjoint element](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

The preimage ideal of the compact operators

Defines the two-sided ideal $J_\rho = \rho^{-1}(\mathcal{K}(H))$.

`MathlibAnnex.CStarAlgebra.compactPreimageIdeal`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Every operator in a singleton image is compact](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A character separating a closed ideal

Finds one character which kills a closed ideal and detects a specified element outside it.

`MathlibAnnex.Analysis.CStarAlgebra.exists_character_annihilating_closedIdeal_of_not_mem`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Every operator in a singleton image is compact](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

An isolated point in a countable Baire space

Applies the Baire property to the closed cover by singletons.

`MathlibAnnex.Topology.exists_isOpen_singleton`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [An isolated character away from the scalar character](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Pure states as real extreme points

Defines purity inside the normalized positive state space.

`MathlibAnnex.Analysis.CStarAlgebra.IsPureState`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Representations without a unit-preservation requirement

Names *-representations of a complex C^* -algebra on a Hilbert space when no unit equation is required.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (6 Cards)

Level 1

A maximal abelian subalgebra containing a self-adjoint element

Starts with the algebra generated by the given element and extends it by inclusion-maximality.

`MathlibAnnex.Analysis.CStarAlgebra.exists_maximalAbelian_containing_isSelfAdjoint`

Immediate Card prerequisites: [Maximal abelian *-subalgebras](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

The GNS representation of a pure state is irreducible

Applies the cyclic pure-state criterion to the space and vector constructed from the same state.

`MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#) · [A character becomes a joint unit eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A character extends to a pure state

Chooses an extreme state in the compact face of all extensions of one prescribed character.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A pure state detects a nonzero square](#) · [A character becomes a joint unit eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Unitary equivalence of two representations

Defines equivalence by one surjective isometry intertwining all algebra actions.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

An isolated character away from the scalar character

Finds an isolated point in the open complement of the distinguished scalar character.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_isolated_character_ne_infinity
```

Immediate Card prerequisites: [An isolated point in a countable Baire space](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Nonzero irreducibility without a unit assumption

Requires nonzero action as well as the absence of proper closed reducing subspaces.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible
```

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2 (2 Cards)

Level 2

A representative of the unique irreducible class

Defines when one nonzero irreducible $\ast$ -representation represents the sole unitary-equivalence class of such representations.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel
```

Immediate Card prerequisites: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#) · [A character becomes a joint unit eigenvector](#)

[Open Declaration Card](#) [Read exact source](#)

Level 2

A pure state detects a nonzero square

Detects a nonzero element through its positive square without an algebra separability assumption.

```
MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_nonzero_on_star_mul_self
```

Immediate Card prerequisites: [A character extends to a pure state](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (2 Cards)

Level 3

A character becomes a joint unit eigenvector

Transfers a normalized GNS eigenvector to the given representation through a unitary with a specified direction.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_unit_eigenvector_of_character_ne_infinity
```

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

Used by in this scope: [The full character space is countable](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

Faithfulness from a unique irreducible class

Uses a pure state on the unitization to detect any hypothetical nonzero element of the kernel.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton
```

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [A pure state detects a nonzero square](#) · [The GNS representation of a pure state is irreducible](#)

Used by in this scope: [A projection represented by a rank-one operator](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (1 Card)

Level 4

The full character space is countable

Uses separated unit eigenvectors to count the non-scalar characters, then restores the one scalar character.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.countable_characterSpace_of_nonUnital_singleton
```

Immediate Card prerequisites: [A character becomes a joint unit eigenvector](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (1 Card)

Level 5

A nonzero projection with scalar corner

Constructs a projection in the original algebra from an isolated non-scalar character of a unitization subalgebra.

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner

Immediate Card prerequisites: [A maximal abelian subalgebra containing a self-adjoint element](#) · [An isolated character away from the scalar character](#) · [The full character space is countable](#)

Used by in this scope: [A projection represented by a rank-one operator](#)

[Open Declaration Card](#) [Read exact source](#)

Level 6 (1 Card)

Level 6

A projection represented by a rank-one operator

Obtains all projection, normalization, image and compactness conditions on the same witnesses.

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_rankOne_map

Immediate Card prerequisites: [Faithfulness from a unique irreducible class](#) · [A nonzero projection with scalar corner](#)

Used by in this scope: [Every rank-one operator has an algebra preimage](#)

[Open Declaration Card](#) [Read exact source](#)

Level 7 (1 Card)

Level 7

Every rank-one operator has an algebra preimage

Turns one represented rank-one projection into every rank-one operator using dense orbits and a closed range.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton`

Immediate Card prerequisites: [A projection represented by a rank-one operator](#)

Used by in this scope: [Every operator in a singleton image is compact](#)

[Open Declaration Card](#) [Read exact source](#)

Level 8 (1 Card)

Level 8

Every operator in a singleton image is compact

Excludes a noncompact image by making one separating character take both zero and one on the same projection.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton`

Immediate Card prerequisites: [Every rank-one operator has an algebra preimage](#) · [A character separating a closed ideal](#) · [The preimage ideal of the compact operators](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)